Probability: Questions and Full Solutions

Question 01

English

A coin is tossed once. If it shows a head, a die is rolled once; if it shows a tail, the coin is tossed two more times. Consider the events

E_1: the die shows an even number,

E_2: at least two tails appear,

E_3: the coin is tossed exactly three times,

E_4: at least one head appears.

Then which of the following statements is TRUE?

(A) E_2 and E_4 are mutually exclusive

(B) E_1 and E_3 are exhaustive

(C) E_3 and E_4 are exhaustive but not mutually exclusive

(D) E_1, E_2 and E_3 are pairwise mutually exclusive

Only one option is correct.

हिन्दी

एक सिक्का एक बार उछाला जाता है। यदि उस पर चित आता है, तो एक पासा एक बार फेंका जाता है; यदि पट आता है, तो सिक्के को दो बार और उछाला जाता है। निम्न घटनाओं पर विचार कीजिए

E_1: पासे पर सम संख्या आती है,

E_2: कम से कम दो पट आते हैं,

E_3: सिक्का ठीक तीन बार उछाला जाता है,

E_4: कम से कम एक चित आता है।

तब निम्न में से कौन-सा कथन सत्य है?

(A) E_2 तथा E_4 परस्पर अपवर्जी हैं

(B) E_1 तथा E_3 निःशेष हैं

(C) E_3 तथा E_4 निःशेष हैं परन्तु परस्पर अपवर्जी नहीं हैं

(D) E_1, E_2 तथा E_3 युग्मतः परस्पर अपवर्जी हैं

केवल एक विकल्प सही है।

Solution

Sample space. A head is followed by one roll of the die, giving the outcomes $H1, H2, \ldots, H6$. A tail is followed by two more tosses, giving THH, THT, TTH, TTT. Hence $S = \{H1, \ldots, H6, THH, THT, TTH, TTT\}$ with $n(S) = 10$ (the outcomes are not equally likely, but the question is only about the sets).

Events as subsets: $E_1 = \{H2, H4, H6\}$. Tails can appear only on the tail branch: THT and TTH have two tails and TTT has three, so $E_2 = \{THT, TTH, TTT\}$. The coin is tossed three times exactly on the tail branch, so $E_3 = \{THH, THT, TTH, TTT\}$. Every outcome except TTT contains a head, so $E_4 = S \setminus \{TTT\}$.

(A) Mutually exclusive means $E_2 \cap E_4 = \phi$. But THT and TTH lie in both, so (A) is wrong.

(B) Exhaustive means $E_1 \cup E_3 = S$. The outcomes $H1, H3, H5$ lie in neither, so (B) is wrong.

(C) The only outcome outside E_4 is TTT, and $TTT \in E_3$; hence $E_3 \cup E_4 = S$, so the events are exhaustive. Also $E_3 \cap E_4 = \{THH, THT, TTH\} \neq \phi$, so they are not mutually exclusive. Hence (C) is correct.

(D) $E_2 \subset E_3$, so $E_2 \cap E_3 = E_2 \neq \phi$; the three events are not pairwise mutually exclusive, so (D) is wrong.

Correct Option: (C)

Question 02

English

Let $S = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ be the sample space of a random experiment, and let a probability be assigned to its elementary events by $P(\{\omega_1\}) = a$, $P(\{\omega_2\}) = 2a$, $P(\{\omega_3\}) = b$ and $P(\{\omega_4\}) = \dfrac{3b^2}{2}$, where a and b are real numbers. If the odds in favour of the event $\{\omega_1, \omega_3\}$ are $1 : 1$, then the odds in favour of the event $\{\omega_4\}$ are

(A) $1 : 5$
(B) $1 : 6$
(C) $5 : 1$
(D) $3 : 1$

Only one option is correct.

हिन्दी

माना कि $S = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ एक यादृच्छिक प्रयोग का प्रतिदर्श समष्टि है, तथा इसकी प्रारम्भिक घटनाओं को प्रायिकताएँ $P(\{\omega_1\}) = a$, $P(\{\omega_2\}) = 2a$, $P(\{\omega_3\}) = b$ तथा $P(\{\omega_4\}) = \dfrac{3b^2}{2}$ द्वारा दी गई हैं, जहाँ a तथा b वास्तविक संख्याएँ हैं। यदि घटना $\{\omega_1, \omega_3\}$ के पक्ष में संयोगानुपात $1 : 1$ है, तो घटना $\{\omega_4\}$ के पक्ष में संयोगानुपात है

(A) $1 : 5$
(B) $1 : 6$
(C) $5 : 1$
(D) $3 : 1$

केवल एक विकल्प सही है।

Solution

Odds in favour of an event E equal to $m : n$ means $P(E) = \dfrac{m}{m+n}$. Hence $P(\{\omega_1, \omega_3\}) = \dfrac{1}{2}$. By additivity over the disjoint elementary events, $a + b = \dfrac{1}{2}$.

Axiom $P(S) = 1$: $a + 2a + b + \dfrac{3b^2}{2} = 1$, i.e. $3a + b + \dfrac{3b^2}{2} = 1$.

Substituting $a = \dfrac{1}{2} - b$: $\dfrac{3}{2} - 3b + b + \dfrac{3b^2}{2} = 1 \Rightarrow \dfrac{3b^2}{2} - 2b + \dfrac{1}{2} = 0 \Rightarrow 3b^2 - 4b + 1 = 0 \Rightarrow (3b-1)(b-1) = 0$.

If $b = 1$, then $a = -\dfrac{1}{2}$, so $P(\{\omega_1\}) < 0$ (and $P(\{\omega_4\}) = \dfrac{3}{2} > 1$). This violates the axiom $P(E) \geq 0$, so this root is rejected.

If $b = \dfrac{1}{3}$, then $a = \dfrac{1}{6}$ and the probabilities are $\dfrac{1}{6}, \dfrac{1}{3}, \dfrac{1}{3}, \dfrac{1}{6}$, all non-negative with sum 1. This is the assignment.

$P(\{\omega_4\}) = \dfrac{3}{2} \cdot \dfrac{1}{9} = \dfrac{1}{6}$ and $P(\{\omega_4\}^c) = \dfrac{5}{6}$, so the odds in favour of $\{\omega_4\}$ are $1 : 5$.

Correct Option: (A)

Question 03

English

A subset X of $\{1, 2, 3, \ldots, 10\}$ is chosen at random, all 2^{10} subsets being equally likely. The probability that each of the three sets $X \cap \{1, 2, 3\}$, $X \cap \{3, 4, 5\}$ and $X \cap \{5, 6, 7\}$ is non-empty is

(A) $\dfrac{39}{128}$

(B) $\dfrac{89}{128}$

(C) $\dfrac{343}{512}$

(D) $\dfrac{85}{128}$

Only one option is correct.

हिन्दी

समुच्चय $\{1, 2, 3, \ldots, 10\}$ का एक उपसमुच्चय X यादृच्छया चुना जाता है, सभी 2^{10} उपसमुच्चय समसम्भाव्य हैं। तीनों समुच्चयों $X \cap \{1, 2, 3\}$, $X \cap \{3, 4, 5\}$ तथा $X \cap \{5, 6, 7\}$ में से प्रत्येक के अरिक्त होने की प्रायिकता है

(A) $\dfrac{39}{128}$

(B) $\dfrac{89}{128}$

(C) $\dfrac{343}{512}$

(D) $\dfrac{85}{128}$

केवल एक विकल्प सही है।

Solution

Since all 2^{10} subsets are equally likely, each element of $\{1, \ldots, 10\}$ belongs to X with probability $\dfrac{1}{2}$, independently of the others. Hence X misses a fixed set of k elements with probability $\dfrac{2^{10-k}}{2^{10}} = 2^{-k}$. Elements $8, 9, 10$ play no role.

Complement: let A_1, A_2, A_3 be the events that X contains no element of $\{1, 2, 3\}$, $\{3, 4, 5\}$, $\{5, 6, 7\}$ respectively. The required event is the complement of 'at least one of A_1, A_2, A_3 occurs'.

Single events: each set has 3 elements, so $P(A_1) = P(A_2) = P(A_3) = \dfrac{1}{8}$.

Pairs: $A_1 \cap A_2$ means X misses $\{1, 2, 3\} \cup \{3, 4, 5\} = \{1, 2, 3, 4, 5\}$ (5 elements, because the sets share 3), probability $\dfrac{1}{32}$. Similarly $P(A_2 \cap A_3) = \dfrac{1}{32}$. But $\{1, 2, 3\}$ and $\{5, 6, 7\}$ are disjoint, so $A_1 \cap A_3$ means missing 6 elements: probability $\dfrac{1}{64}$.

Triple: X misses $\{1, \ldots, 7\}$, probability $\dfrac{1}{128}$.

Addition theorem for three events: $P(A_1 \cup A_2 \cup A_3) = \dfrac{3}{8} - \dfrac{1}{32} - \dfrac{1}{32} - \dfrac{1}{64} + \dfrac{1}{128} = \dfrac{48 - 4 - 4 - 2 + 1}{128} = \dfrac{39}{128}$.

Hence the required probability is $1 - \dfrac{39}{128} = \dfrac{89}{128}$. (Treating the three conditions as independent would give $\left(\dfrac{7}{8}\right)^3 = \dfrac{343}{512}$, which is wrong because the sets overlap.)

Correct Option: (B)

Question 04

English

A biased coin, for which the probability of a head in a single toss is $\frac{1}{3}$, is tossed 6 times. Given that at least 4 heads are obtained, the conditional probability that the first toss shows a head is

(A) $\frac{1}{3}$

(B) $\frac{2}{3}$

(C) $\frac{61}{73}$

(D) $\frac{51}{73}$

Only one option is correct.

हिन्दी

एक अभिनत सिक्का, जिसके एक उछाल में चित आने की प्रायिकता $\frac{1}{3}$ है, 6 बार उछाला जाता है। दिया है कि कम से कम 4 चित प्राप्त होते हैं, तो पहली उछाल में चित आने की सप्रतिबन्ध प्रायिकता है

(A) $\frac{1}{3}$

(B) $\frac{2}{3}$

(C) $\frac{61}{73}$

(D) $\frac{51}{73}$

केवल एक विकल्प सही है।

Solution

Let X be the number of heads; $X \sim B\left(6, \frac{1}{3}\right)$ with $p = \frac{1}{3}$, $q = \frac{2}{3}$.

$P(X = 4) = \binom{6}{4}p^4q^2 = \frac{15 \cdot 4}{729} = \frac{60}{729}$, $P(X = 5) = \binom{6}{5}p^5q = \frac{12}{729}$, $P(X = 6) = \frac{1}{729}$. Hence $P(X \geq 4) = \frac{73}{729}$.

Reduced sample space: given $X = k$, each of the $\binom{6}{k}$ head-tail sequences has the same probability p^kq^{6-k}, so they are equally likely. The first toss is a head in $\binom{5}{k-1}$ of them, and $\frac{\binom{5}{k-1}}{\binom{6}{k}} = \frac{k}{6}$.

Therefore $P(\text{first H} \cap X \geq 4) = \frac{4}{6} \cdot \frac{60}{729} + \frac{5}{6} \cdot \frac{12}{729} + \frac{6}{6} \cdot \frac{1}{729} = \frac{40 + 10 + 1}{729} = \frac{51}{729}$.

Check by direct counting: first toss a head and at least 3 heads among the remaining 5 tosses: $p\left[\binom{5}{3}p^3q^2 + \binom{5}{4}p^4q + p^5\right] = \frac{1}{3} \cdot \frac{40 + 10 + 1}{243} = \frac{51}{729}$.

$P(\text{first H} \mid X \geq 4) = \frac{51/729}{73/729} = \frac{51}{73}$. The value $\frac{1}{3}$ would hold only without the condition.

Correct Option: (D)

Question 05

English

In a sequence of n independent trials, each trial results in a success with the same probability p, where $0 < p < 1$. Let X denote the number of successes, and suppose that the mean and the variance of X are 4 and 3 respectively. Let N denote the number of ordered pairs (i, j) with $1 \leq i, j \leq n$ such that the i-th trial is a success and the j-th trial is a failure. Then the mean of N is

(A) 45
(B) 48
(C) 42
(D) 51

Only one option is correct.

हिन्दी

n स्वतन्त्र परीक्षणों के एक अनुक्रम में प्रत्येक परीक्षण समान प्रायिकता p से सफल होता है, जहाँ $0 < p < 1$ है। माना कि X सफलताओं की संख्या है, तथा X का माध्य तथा प्रसरण क्रमशः 4 तथा 3 हैं। माना कि N उन क्रमित युग्मों (i, j) की संख्या है, जहाँ $1 \leq i, j \leq n$ है, जिनके लिए i-वाँ परीक्षण सफल तथा j-वाँ परीक्षण असफल होता है। तब N का माध्य है

(A) 45
(B) 48
(C) 42
(D) 51

केवल एक विकल्प सही है।

Solution

For a binomial variable, mean $np = 4$ and variance $npq = 3$. Dividing, $q = \dfrac{3}{4}$, so $p = \dfrac{1}{4}$ and $n = \dfrac{4}{p} = 16$.

Counting N: a pair (i, j) is counted exactly when trial i is one of the X successes and trial j is one of the $16 - X$ failures (then automatically $i \neq j$). Hence $N = X(16 - X)$, a function of the discrete random variable X.

Mean of a function of X: $E(N) = E(16X - X^2) = 16E(X) - E(X^2)$.

From the definition of variance, $E(X^2) = \mathrm{Var}(X) + [E(X)]^2 = 3 + 16 = 19$.

So $E(N) = 16 \cdot 4 - 19 = 45$.

Independent check by linearity: each ordered pair (i, j) with $i \neq j$ contributes with probability pq, and there are $16 \cdot 15$ such pairs: $240 \cdot \dfrac{1}{4} \cdot \dfrac{3}{4} = 45$. (Using $E(X^2) = [E(X)]^2$ would give 48, a typical error.)

Correct Option: (A)

Question 06

English

For every subset X of the set $\{1, 2, 3, \ldots, 12\}$, including the empty set and the whole set, the observation $d_X = n(X) - n(X')$ is recorded, where X' denotes the complement of X in $\{1, 2, 3, \ldots, 12\}$. Then the standard deviation of these 2^{12} observations is

(A) $\sqrt{3}$
(B) $2\sqrt{3}$
(C) 12
(D) 6

Only one option is correct.

हिन्दी

समुच्चय $\{1, 2, 3, \ldots, 12\}$ के प्रत्येक उपसमुच्चय X (रिक्त समुच्चय तथा पूर्ण समुच्चय सहित) के लिए प्रेक्षण $d_X = n(X) - n(X')$ लिखा जाता है, जहाँ X' समुच्चय $\{1, 2, 3, \ldots, 12\}$ में X का पूरक है। तब इन 2^{12} प्रेक्षणों का मानक विचलन है

(A) $\sqrt{3}$
(B) $2\sqrt{3}$
(C) 12
(D) 6

केवल एक विकल्प सही है।

Solution

Since $n(X') = 12 - n(X)$, each observation is $d_X = 2n(X) - 12$.

Frequency distribution of $n(X)$: exactly $\binom{12}{k}$ subsets have k elements, $k = 0, 1, \ldots, 12$, and the total frequency is 2^{12}.

The relative frequency of k is $\frac{\binom{12}{k}}{2^{12}} = \binom{12}{k}\left(\frac{1}{2}\right)^k\left(\frac{1}{2}\right)^{12-k}$, which is the binomial distribution $B\left(12, \frac{1}{2}\right)$. The mean and variance of the data equal the mean and variance of this distribution: $\bar{x} = np = 6$, $\sigma^2 = npq = 3$.

Effect of a linear change on ungrouped data: if $y_i = ax_i + b$ then $\sigma_y^2 = a^2\sigma_x^2$. With $a = 2$, $b = -12$: $\sigma_d^2 = 4 \cdot 3 = 12$, and the mean of d is $2 \cdot 6 - 12 = 0$.

Hence $\sigma_d = \sqrt{12} = 2\sqrt{3}$. (Option (C) is the variance and option (A) is the standard deviation of $n(X)$, both typical slips.)

Correct Option: (B)

Question 07

English

Three fair coins are tossed simultaneously, and S denotes the sample space of this random experiment. Let A be an event (a subset of S) such that the odds in favour of A are $5 : 3$. Let a be the number of events $B \subseteq S$ such that A and B are mutually exclusive, and let b be the number of events $B \subseteq S$ such that A and B are exhaustive, that is, $A \cup B = S$. Then the value of $a - b$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

तीन अनभिनत सिक्के एक साथ उछाले जाते हैं, तथा S इस यादृच्छिक प्रयोग का प्रतिदर्श समष्टि है। माना कि A एक घटना (S का उपसमुच्चय) है जिसके पक्ष में संयोगानुपात $5 : 3$ है। माना कि a उन घटनाओं $B \subseteq S$ की संख्या है जिनके लिए A तथा B परस्पर अपवर्जी हैं, तथा b उन घटनाओं $B \subseteq S$ की संख्या है जिनके लिए A तथा B निःशेष हैं, अर्थात् $A \cup B = S$ है। तब $a - b$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Sample space: S consists of the $2^3 = 8$ ordered outcomes $HHH, HHT, \ldots, TTT$, all equally likely, so $P(E) = \dfrac{n(E)}{8}$ for every event E.

Odds in favour of A are $5 : 3$, so $P(A) = \dfrac{5}{5+3} = \dfrac{5}{8}$ and therefore $n(A) = 5$, $n(A') = 3$.

Counting a: A and B are mutually exclusive exactly when $A \cap B = \phi$, i.e. $B \subseteq A'$. Every subset of the 3-element set A' works, including ϕ: $a = 2^3 = 8$.

Counting b: A and B are exhaustive exactly when $A \cup B = S$, i.e. B contains every outcome of A'. The 5 outcomes of A may or may not be in B: $b = 2^5 = 32$.

Hence $a - b = 8 - 32 = -24$. The answer does not depend on which 5 outcomes form A.

Final Answer: -24

Question 08

English

An integer n is chosen at random from $\{1, 2, 3, \ldots, 100\}$. Let p be the probability that n is divisible by at least one of the numbers 6, 10 and 15. Then the value of $100p$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

समुच्चय $\{1, 2, 3, \ldots, 100\}$ से एक पूर्णांक n यादृच्छया चुना जाता है। माना कि p इसकी प्रायिकता है कि n संख्याओं 6, 10 तथा 15 में से कम से कम एक से विभाज्य है। तब $100p$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let A, B, C be the events that n is divisible by 6, 10, 15 respectively. All 100 choices are equally likely, so $p = \dfrac{n(A \cup B \cup C)}{100}$.

$n(A) = \left[\dfrac{100}{6}\right] = 16$, $n(B) = \left[\dfrac{100}{10}\right] = 10$, $n(C) = \left[\dfrac{100}{15}\right] = 6$.

Divisible by two of them means divisible by their lcm: $\text{lcm}(6, 10) = 30$, $\text{lcm}(10, 15) = 30$, $\text{lcm}(6, 15) = 30$. Divisible by all three: $\text{lcm}(6, 10, 15) = 30$. So each pairwise intersection and the triple intersection is the set $\{30, 60, 90\}$ with 3 elements.

Inclusion-exclusion (addition theorem for three events): $n(A \cup B \cup C) = 16+10+6-(3+3+3)+3 = 26$.

Check by direct listing: the 16 multiples of 6, plus the multiples of 10 not divisible by 6 $(10, 20, 40, 50, 70, 80, 100$: 7), plus the multiples of 15 divisible by neither $(15, 45, 75$: 3): 26.

Hence $p = 0.26$ and $100p = 26$.

Final Answer: 26

Question 09

English

Two fair dice are rolled, and a and b denote the numbers shown on them. Let p be the probability that the product ab is divisible by the sum $a + b$, and let q be the probability that a and b are coprime. Then the value of $p - q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

दो अनभिनत पासे फेंके जाते हैं, तथा a और b उन पर आई संख्याएँ हैं। माना कि p इसकी प्रायिकता है कि गुणनफल ab योग $a + b$ से विभाज्य है, तथा q इसकी प्रायिकता है कि a और b सहअभाज्य हैं। तब $p - q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The 36 ordered pairs (a, b) are equally likely.

Key observation for p: if $\gcd(a, b) = 1$, any prime dividing $a + b$ and ab divides a or b, and then also the other one; so $\gcd(ab, a + b) = 1$. Since $a + b \geq 2$, $a + b$ cannot divide ab. Hence only pairs with a common factor need checking.

Pairs with $\gcd(a, b) > 1$ (unordered): $(2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (4, 6), (5, 5), (6, 6)$.

$(2, 2)$: $4 \mid 4$ yes. $(2, 4)$: $6 \nmid 8$. $(2, 6)$: $8 \nmid 12$. $(3, 3)$: $6 \nmid 9$. $(3, 6)$: $9 \mid 18$ yes. $(4, 4)$: $8 \mid 16$ yes. $(4, 6)$: $10 \nmid 24$. $(5, 5)$: $10 \nmid 25$. $(6, 6)$: $12 \mid 36$ yes.

Favourable ordered pairs: $(2, 2), (4, 4), (6, 6), (3, 6), (6, 3)$, so $p = \dfrac{5}{36}$.

For q, count coprime ordered pairs row by row: $a = 1$: all 6; $a = 2$: $b = 1, 3, 5$; $a = 3$: $b = 1, 2, 4, 5$; $a = 4$: $b = 1, 3, 5$; $a = 5$: $b = 1, 2, 3, 4, 6$; $a = 6$: $b = 1, 5$. Total $6 + 3 + 4 + 3 + 5 + 2 = 23$, so $q = \dfrac{23}{36}$.

$p - q = \dfrac{5 - 23}{36} = -\dfrac{18}{36} = -0.50$.

Final Answer: -0.50

Question 10

English

The ten cards ace, $2, 3, \ldots, 10$ of hearts are taken from a pack of playing cards, the ace being counted as 1. Three of these ten cards are drawn at random, simultaneously. Then the probability that the sum of the numbers on the three cards drawn is divisible by 3 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

ताश की एक गड्डी से पान के दस पत्ते इक्का, $2, 3, \ldots, 10$ लिए जाते हैं, जहाँ इक्के को 1 गिना जाता है। इन दस पत्तों में से तीन पत्ते एक साथ यादृच्छया निकाले जाते हैं। तब निकाले गए तीनों पत्तों पर अंकित संख्याओं का योग 3 से विभाज्य होने की प्रायिकता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

All $\binom{10}{3} = 120$ selections of three cards are equally likely.

Only the remainders on division by 3 matter. Remainder 0: $3, 6, 9$ (3 cards). Remainder 1: $1, 4, 7, 10$ (4 cards). Remainder 2: $2, 5, 8$ (3 cards).

Three remainders $r_1, r_2, r_3 \in \{0, 1, 2\}$ add to a multiple of 3 exactly when they are all equal ($0+0+0$, $1+1+1$, $2+2+2$) or all different ($0+1+2$); two equal and one different never works.

Counts: all remainder 0: $\binom{3}{3} = 1$; all remainder 1: $\binom{4}{3} = 4$; all remainder 2: $\binom{3}{3} = 1$; one of each: $3 \cdot 4 \cdot 3 = 36$. Total 42.

Probability $= \dfrac{42}{120} = \dfrac{7}{20} = 0.35$. (The value $\dfrac{1}{3}$ would hold only if the three classes had equal sizes.)

Final Answer: 0.35

Question 11

English

A four-digit number is formed by choosing each of its digits from $\{1, 2, 3, 4, 5\}$, repetition of digits being allowed, all such numbers being equally likely. Let p be the probability that the number formed is greater than the number obtained by writing its digits in the reverse order. Then the value of $625p$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक चार अंकों की संख्या इस प्रकार बनाई जाती है कि उसका प्रत्येक अंक $\{1, 2, 3, 4, 5\}$ से चुना जाता है, अंकों की पुनरावृत्ति की अनुमति है, तथा ऐसी सभी संख्याएँ समसम्भाव्य हैं। माना कि p इसकी प्रायिकता है कि बनी हुई संख्या, उसके अंकों को उल्टे क्रम में लिखने से प्राप्त संख्या से बड़ी है। तब $625p$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Formation of numbers with repetition: each of the 4 places can be filled in 5 ways, giving $5^4 = 625$ equally likely numbers. No digit is 0, so the reverse of each is again one of these numbers.

Let $\overline{N}$ denote the reverse of $N = d_1d_2d_3d_4$. The map $N \mapsto \overline{N}$ is a bijection of the 625 numbers onto themselves and $\overline{\overline{N}} = N$.

$N = \overline{N}$ exactly when $d_1 = d_4$ and $d_2 = d_3$; choosing d_1 and d_2 gives $5 \cdot 5 = 25$ such numbers.

The remaining $625 - 25 = 600$ numbers fall into 300 pairs $\{N, \overline{N}\}$ with $N \neq \overline{N}$, and in each pair exactly one number is the larger. So exactly 300 numbers are greater than their reverse.

$p = \dfrac{300}{625} = \dfrac{12}{25}$ and $625p = 300$.

Final Answer: 300

Question 12

English

All the letters of the word RESIST are arranged at random, all the distinct arrangements being equally likely, and the arrangement obtained is regarded as a word W. Let p be the probability that, in dictionary order, W comes after the word RESIST and before the word SISTER. Then the value of $\frac{1}{p}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

शब्द RESIST के सभी अक्षरों को यादृच्छया व्यवस्थित किया जाता है, सभी भिन्न व्यवस्थाएँ समसम्भाव्य हैं, तथा प्राप्त व्यवस्था को एक शब्द W माना जाता है। माना कि p इसकी प्रायिकता है कि शब्दकोश क्रम में W शब्द RESIST के बाद तथा शब्द SISTER से पहले आता है। तब $\frac{1}{p}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Sample space: the letters E, I, R, S, S, T (with S twice) give $\frac{6!}{2!} = 360$ distinct words, all equally likely. RESIST and SISTER are two of them, so $p = \frac{r_2 - r_1 - 1}{360}$, where r_1, r_2 are their ranks in dictionary order.

Alphabetical order of the letters: $E < I < R < S < T$.

Rank of RESIST. Words beginning with E: the other letters I, R, S, S, T give $\frac{5!}{2!} = 60$. Beginning with I: also 60. Beginning with R: second letter E is the smallest, so we continue with RE and the letters I, S, S, T. Words $REI\ldots$: $\frac{3!}{2!} = 3$. Then RES with I, S, T left: the first such word is $RESIST$ itself. So $r_1 = 60 + 60 + 3 + 1 = 124$.

Rank of SISTER. Words beginning with E, I, R: 60 each, total 180. Beginning with S (letters E, I, R, S, T left, all distinct): $SE\ldots$: $4! = 24$. Then SI with E, R, S, T left: $SIE\ldots$: $3! = 6$, $SIR\ldots$: 6. Then SIS with E, R, T left: $SISE\ldots$: 2, $SISR\ldots$: 2; then $SIST$ with E, R: $SISTER$ is the first. So $r_2 = 180 + 24 + 6 + 6 + 2 + 2 + 1 = 221$.

Number of words strictly between: $221 - 124 - 1 = 96$. Hence $p = \frac{96}{360} = \frac{4}{15}$.

$\frac{1}{p} = \frac{15}{4} = 3.75$.

Final Answer: 3.75

Question 13

English

Five distinct balls are placed into four distinct boxes, each ball being put into any one of the four boxes with equal probability, independently of the other balls. Let p be the probability that no box is empty, and let q be the probability that exactly one box is empty. Then the value of $\frac{q}{p}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

पाँच भिन्न गेंदें चार भिन्न डिब्बों में रखी जाती हैं, प्रत्येक गेंद अन्य गेंदों से स्वतन्त्र रूप से चारों डिब्बों में से किसी एक में समान प्रायिकता से रखी जाती है। माना कि p इसकी प्रायिकता है कि कोई भी डिब्बा खाली नहीं है, तथा q इसकी प्रायिकता है कि ठीक एक डिब्बा खाली है। तब $\frac{q}{p}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Each ball has 4 choices, so there are $4^5 = 1024$ equally likely placements; p and q are counts divided by 1024, so $\frac{q}{p}$ is a ratio of counts.

No box empty. Let B_i be the set of placements with box i empty. Placements with k specified boxes empty: $(4-k)^5$. By inclusion-exclusion the number with no empty box is $4^5 - \binom{4}{1}3^5 + \binom{4}{2}2^5 - \binom{4}{3}1^5 = 1024 - 972 + 192 - 4 = 240$. (Check: the occupancy pattern is $2,1,1,1$: $4 \cdot \binom{5}{2} \cdot 3! = 240$.)

Exactly one box empty. Choose the empty box in 4 ways. The 5 balls must then go into the remaining 3 boxes with none of these empty: $3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 243 - 96 + 3 = 150$. Total $4 \cdot 150 = 600$.

Hence $p = \frac{240}{1024}$, $q = \frac{600}{1024}$ and $\frac{q}{p} = \frac{600}{240} = \frac{5}{2} = 2.5$.

Final Answer: 2.50

Question 14

English

An ordered triple (x, y, z) is chosen at random from the set of all ordered triples of non-negative integers satisfying $x + y + z = 23$, all such triples being equally likely. Let p be the probability that each of x, y, z is at most 11, and let q be the probability that each of x, y, z is at least 1. Then the value of $p - q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

समीकरण $x + y + z = 23$ को सन्तुष्ट करने वाले अऋणात्मक पूर्णांकों के सभी क्रमित त्रिकों के समुच्चय से एक क्रमित त्रिक (x, y, z) यादृच्छया चुना जाता है, ऐसे सभी त्रिक समसम्भाव्य हैं। माना कि p इसकी प्रायिकता है कि x, y, z में से प्रत्येक अधिकतम 11 है, तथा q इसकी प्रायिकता है कि x, y, z में से प्रत्येक कम से कम 1 है। तब $p - q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Sample space: the number of non-negative integer solutions of $x+y+z = 23$ is $\binom{23+3-1}{3-1} = \binom{25}{2} = 300$; all are equally likely.

Lower bounds (q): with $x = x' + 1$, $y = y' + 1$, $z = z' + 1$ and $x', y', z' \geq 0$, the equation becomes $x' + y' + z' = 20$, which has $\binom{22}{2} = 231$ solutions. So $q = \dfrac{231}{300} = 0.77$.

Upper bounds (p): let A_x be the solutions with $x \geq 12$. Putting $x = x'' + 12$ gives $x'' + y + z = 11$: $\binom{13}{2} = 78$ solutions. The same holds for A_y, A_z. Two variables cannot both be at least 12 because their sum would exceed 23, so the sets are disjoint.

Hence the number with every variable at most 11 is $300 - 3 \cdot 78 = 66$ and $p = \dfrac{66}{300} = 0.22$.

$p - q = 0.22 - 0.77 = -0.55$.

Final Answer: -0.55

Question 15

English

Let A, B and C be three mutually independent events such that $P(A \cup B) = 0.8$, $P(B \cup C) = 0.7$ and $P(C \cup A) = 0.76$. Then the value of $100\,P(A \cup B \cup C)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि A, B तथा C तीन परस्पर स्वतन्त्र घटनाएँ हैं जिनके लिए $P(A \cup B) = 0.8$, $P(B \cup C) = 0.7$ तथा $P(C \cup A) = 0.76$ है। तब $100\,P(A \cup B \cup C)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Independence of A, B, C gives independence of A', B', C'. Let $\alpha = P(A')$, $\beta = P(B')$, $\gamma = P(C')$.

Complement of 'at least one': $P(A \cup B) = 1 - P(A' \cap B') = 1 - \alpha\beta$. (Equivalently, by the addition theorem, $P(A \cup B) = P(A) + P(B) - P(A)P(B) = 1 - (1 - P(A))(1 - P(B))$.)

Hence $\alpha\beta = 0.2$, $\beta\gamma = 0.3$, $\gamma\alpha = 0.24$.

Multiplying: $(\alpha\beta\gamma)^2 = 0.2 \cdot 0.3 \cdot 0.24 = 0.0144$, and since all three are probabilities, $\alpha\beta\gamma = 0.12$ (the root -0.12 is impossible).

Then $\gamma = \dfrac{0.12}{0.2} = 0.6$, $\alpha = \dfrac{0.12}{0.3} = 0.4$, $\beta = \dfrac{0.12}{0.24} = 0.5$. So $P(A) = 0.6$, $P(B) = 0.5$, $P(C) = 0.4$, and the data are consistent: $1 - 0.4 \cdot 0.5 = 0.8$, $1 - 0.5 \cdot 0.6 = 0.7$, $1 - 0.6 \cdot 0.4 = 0.76$.

At least one of the three: $P(A \cup B \cup C) = 1 - P(A' \cap B' \cap C') = 1 - \alpha\beta\gamma = 0.88$. Check with the addition theorem: $1.5 - (0.3 + 0.2 + 0.24) + 0.12 = 0.88$.

Hence $100P(A \cup B \cup C) = 88$.

Final Answer: 88

Question 16

English

Consider n points in a plane, $n \geq 4$, no three of which are collinear, and draw the $\binom{n}{2}$ lines each joining two of these points. Assume that no two of these lines are parallel and that no three of them pass through a common point other than one of the n given points. Two of these lines are chosen at random. It is given that the probability that their point of intersection is not one of the n given points is $\frac{7}{11}$. Then the number of points of intersection of these lines, other than the n given points, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक तल में n बिन्दुओं $(n \geq 4)$ पर विचार कीजिए, जिनमें से कोई तीन संरेख नहीं हैं, तथा इनमें से दो-दो बिन्दुओं को मिलाने वाली $\binom{n}{2}$ रेखाएँ खींचिए। मान लीजिए कि इन रेखाओं में से कोई दो समान्तर नहीं हैं तथा n दिए गए बिन्दुओं के अतिरिक्त किसी अन्य बिन्दु से इनमें से कोई तीन रेखाएँ नहीं गुज़रतीं। इन रेखाओं में से दो रेखाएँ यादृच्छया चुनी जाती हैं। दिया है कि उनका प्रतिच्छेदन बिन्दु n दिए गए बिन्दुओं में से कोई न होने की प्रायिकता $\frac{7}{11}$ है। तब इन रेखाओं के, n दिए गए बिन्दुओं के अतिरिक्त, प्रतिच्छेदन बिन्दुओं की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Lines from n points with no three collinear: $L = \binom{n}{2} = \dfrac{n(n-1)}{2}$, and the number of pairs of lines is $\binom{L}{2} = \dfrac{L(L-1)}{2} = \dfrac{n(n-1)}{2} \cdot \dfrac{n^2-n-2}{4} = \dfrac{n(n-1)(n-2)(n+1)}{8}$.

Two distinct lines (not parallel) meet in exactly one point. Their intersection is a given point exactly when the two lines have a given point in common. For each of the n points, the lines through it are the $n-1$ lines joining it to the others, giving $\binom{n-1}{2}$ pairs; two distinct lines share at most one given point, so there is no double counting: $n\binom{n-1}{2} = \dfrac{n(n-1)(n-2)}{2}$ pairs.

So $P(\text{intersection is a given point}) = \dfrac{n(n-1)(n-2)/2}{n(n-1)(n-2)(n+1)/8} = \dfrac{4}{n+1}$, and by the complement rule $P(\text{not a given point}) = \dfrac{n-3}{n+1}$.

$\dfrac{n-3}{n+1} = \dfrac{7}{11} \Rightarrow 11n - 33 = 7n + 7 \Rightarrow n = 10.$

For $n = 10$: $L = 45$ lines and $\binom{45}{2} = 990$ pairs, of which $10\binom{9}{2} = 360$ meet at given points. The other 630 pairs meet at points other than the given ones, and since no three lines are concurrent there, different pairs give different points. (Check: every 4 of the points give 3 such points, $3\binom{10}{4} = 630$.)

Final Answer: 630

Question 17

English

A fair die is rolled once, and b denotes the number shown. Then b fair coins are tossed, and c denotes the number of heads obtained. Consider the quadratic equation $x^2 + bx + c = 0$. Given that this equation has two distinct integer roots, let k be the integer for which the conditional probability that the smaller root of the equation equals k is the greatest. Then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक अनभिनत पासा एक बार फेंका जाता है, तथा b उस पर आई संख्या है। फिर b अनभिनत सिक्के उछाले जाते हैं, तथा c प्राप्त चितों की संख्या है। द्विघात समीकरण $x^2 + bx + c = 0$ पर विचार कीजिए। दिया है कि इस समीकरण के दो भिन्न पूर्णांक मूल हैं; माना कि k वह पूर्णांक है जिसके लिए समीकरण के छोटे मूल के k के बराबर होने की सप्रतिबन्ध प्रायिकता सबसे अधिक है। तब k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Joint distribution by the multiplication theorem: $P(b = \beta) = \dfrac{1}{6}$ and, given $b = \beta$, c is binomial, so $P(b = \beta, c = \gamma) = P(b = \beta)\, P(c = \gamma \mid b = \beta) = \dfrac{1}{6} \cdot \dfrac{\binom{\beta}{\gamma}}{2^\beta}$ for $0 \le \gamma \le \beta$.

Event E: two distinct integer roots. Since $b, c \ge 0$ and $b > 0$, the roots are $-u, -v$ with integers $u, v \ge 0$, $u \ne v$, $u + v = b$, $uv = c \le b$.

If $c = 0$: roots 0 and $-b$, distinct for every $b = 1, \ldots, 6$. If $c \ge 1$: $u, v \ge 1$ and $uv \le u + v$ forces $\min(u, v) = 1$ (since $(u - 1)(v - 1) \le 1$ with $u \ne v$), giving $(u, v) = (1, b - 1)$ with $b - 1 \ge 2$: $(b, c) = (3, 2), (4, 3), (5, 4), (6, 5)$.

Multiply all probabilities by $384 = 6 \cdot 64$: $(1, 0)$: 32, $(2, 0)$: 16, $(3, 0)$: 8, $(4, 0)$: 4, $(5, 0)$: 2, $(6, 0)$: 1, $(3, 2)$: $\binom{3}{2} \cdot 8 = 24$, $(4, 3)$: $\binom{4}{3} \cdot 4 = 16$, $(5, 4)$: $\binom{5}{4} \cdot 2 = 10$, $(6, 5)$: $\binom{6}{5} \cdot 1 = 6$. So $P(E) = \dfrac{119}{384}$.

Smaller root: for $(b, 0)$ it is $-b$; for $(b, b - 1)$ it is $-(b - 1)$. Grouping: -1: 32; -2: $16 + 24 = 40$; -3: $8 + 16 = 24$; -4: $4 + 10 = 14$; -5: $2 + 6 = 8$; -6: 1.

Conditional probabilities on the reduced sample space E: $P(\text{smaller root} = k \mid E) = \dfrac{w_k}{119}$, greatest for $k = -2$ with $\dfrac{40}{119}$.

Final Answer: -2

Question 18

English

A fair die is rolled, and if it shows the number k, an urn is filled with k white balls and $6-k$ black balls. Three balls are then drawn from the urn one after another without replacement. Given that the first two balls drawn are white, the conditional probability that the third ball drawn is also white is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक अनभिनत पासा फेंका जाता है, तथा यदि उस पर संख्या k आती है, तो एक कलश में k सफेद गेंदें तथा $6-k$ काली गेंदें भरी जाती हैं। फिर कलश से तीन गेंदें एक के बाद एक बिना प्रतिस्थापन के निकाली जाती हैं। दिया है कि पहली दो निकाली गई गेंदें सफेद हैं, तो तीसरी निकाली गई गेंद के भी सफेद होने की सप्रतिबन्ध प्रायिकता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let K be the number on the die and W_i the event that the i-th ball is white. Given $K=k$, by the multiplication theorem for draws without replacement: $P(W_1 \cap W_2 \mid K=k) = \dfrac{k}{6}\cdot\dfrac{k-1}{5} = \dfrac{k(k-1)}{30}$ and $P(W_1 \cap W_2 \cap W_3 \mid K=k) = \dfrac{k(k-1)(k-2)}{120}$.

Law of total probability over $k=1,\dots,6$: $P(W_1\cap W_2) = \dfrac{1}{6}\cdot\dfrac{0+2+6+12+20+30}{30} = \dfrac{70}{180} = \dfrac{7}{18}$ and $P(W_1\cap W_2\cap W_3) = \dfrac{1}{6}\cdot\dfrac{0+0+6+24+60+120}{120} = \dfrac{210}{720} = \dfrac{7}{24}$.

Conditional probability: $P(W_3 \mid W_1\cap W_2) = \dfrac{P(W_1\cap W_2\cap W_3)}{P(W_1\cap W_2)} = \dfrac{7/24}{7/18} = \dfrac{18}{24} = \dfrac{3}{4}$.

The same through Bayes' theorem: $P(K=k \mid W_1\cap W_2) = \dfrac{k(k-1)}{70}$ (so the urn with 6 white balls has posterior $\dfrac{30}{70}$); then $P(W_3 \mid W_1\cap W_2) = \sum_k \dfrac{k(k-1)}{70}\cdot\dfrac{k-2}{4} = \dfrac{210}{280} = 0.75$.

Final Answer: 0.75

Question 19

English

An urn contains 3 red balls and 2 green balls. A ball is drawn at random; if it is red, it is kept aside, and if it is green, it is returned to the urn together with two more green balls. A second ball is then drawn at random from the urn. A player wins Rs. 70 if the second ball drawn is red and loses Rs. 70 if it is green. Then the expected gain, in rupees, of the player is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक कलश में 3 लाल तथा 2 हरी गेंदें हैं। एक गेंद यादृच्छया निकाली जाती है; यदि वह लाल है तो उसे अलग रख दिया जाता है, और यदि वह हरी है तो उसे दो और हरी गेंदों के साथ कलश में वापस डाल दिया जाता है। फिर कलश से दूसरी गेंद यादृच्छया निकाली जाती है। यदि दूसरी निकाली गई गेंद लाल है तो खिलाड़ी 70 रुपये जीतता है तथा यदि हरी है तो 70 रुपये हारता है। तब खिलाड़ी का प्रत्याशित लाभ (रुपयों में) _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let R_1, G_1 be the colour of the first ball and R_2 the event that the second ball is red. $P(R_1) = \dfrac{3}{5}$, $P(G_1) = \dfrac{2}{5}$.

Reduced urns: given R_1, the red ball is kept aside, leaving 2 red and 2 green, so $P(R_2 \mid R_1) = \dfrac{1}{2}$.

Given G_1, the urn becomes 3 red and 4 green, so $P(R_2 \mid G_1) = \dfrac{3}{7}$.

Multiplication theorem: $P(R_1 \cap R_2) = \dfrac{3}{5} \cdot \dfrac{1}{2} = \dfrac{3}{10}$, $P(G_1 \cap R_2) = \dfrac{2}{5} \cdot \dfrac{3}{7} = \dfrac{6}{35}$.

Law of total probability: $P(R_2) = \dfrac{3}{10} + \dfrac{6}{35} = \dfrac{21+12}{70} = \dfrac{33}{70}$, so $P(\text{second green}) = \dfrac{37}{70}$.

Expected gain $= 70 \cdot \dfrac{33}{70} + (-70) \cdot \dfrac{37}{70} = 33 - 37 = -4$ rupees.

Final Answer: -4

Question 20

English

Let A, B and C be three events such that $P(A) = 0.5$, $P(B) = 0.4$ and $P(C) = 0.6$, each pair of the three events is independent, and the probability that none of A, B, C occurs is 0.15. Then the value of $P(A \cap B \cap C) - P(A)P(B)P(C)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि A, B तथा C तीन घटनाएँ हैं जिनके लिए $P(A) = 0.5$, $P(B) = 0.4$ तथा $P(C) = 0.6$ है, इन तीनों में से प्रत्येक युग्म स्वतन्त्र है, तथा A, B, C में से किसी के भी घटित न होने की प्रायिकता 0.15 है। तब $P(A \cap B \cap C) - P(A)P(B)P(C)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For independent events the multiplication theorem gives $P(E \cap F) = P(E)P(F)$. Pairwise independence therefore gives $P(A \cap B) = 0.5 \cdot 0.4 = 0.2$, $P(A \cap C) = 0.5 \cdot 0.6 = 0.3$, $P(B \cap C) = 0.4 \cdot 0.6 = 0.24$. It says nothing about $P(A \cap B \cap C)$.

The probability that at least one occurs is $1 - 0.15 = 0.85$. The addition theorem for three events gives $0.85 = (0.5 + 0.4 + 0.6) - (0.2 + 0.3 + 0.24) + P(A \cap B \cap C) = 0.76 + P(A \cap B \cap C)$, so $P(A \cap B \cap C) = 0.09$.

The data are consistent: the regions 'all three' 0.09, 'exactly A, B' 0.11, 'exactly A, C' 0.21, 'exactly B, C' 0.15, 'only A' 0.09, 'only B' 0.05, 'only C' 0.15, 'none' 0.15 are non-negative and add up to 1.

Mutual independence would need $P(A \cap B \cap C) = P(A)P(B)P(C) = 0.12$; here $0.09 \neq 0.12$, so the events are pairwise but not mutually independent.

$P(A \cap B \cap C) - P(A)P(B)P(C) = 0.09 - 0.12 = -0.03$.

Final Answer: -0.03

Question 21

English

In one trial, two fair dice are rolled. Let A be the event that the first die shows an odd number, B the event that the second die shows an odd number, and C the event that the sum of the two numbers is odd. Five such trials are performed independently. Let p be the probability that none of the events A, B, C occurs in exactly two of the five trials. Then the value of $1024p$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक परीक्षण में दो अनभिनत पासे फेंके जाते हैं। माना कि A घटना है कि पहले पासे पर विषम संख्या आती है, B घटना है कि दूसरे पासे पर विषम संख्या आती है, तथा C घटना है कि दोनों संख्याओं का योग विषम है। ऐसे पाँच परीक्षण स्वतन्त्र रूप से किए जाते हैं। माना कि p इसकी प्रायिकता है कि पाँच में से ठीक दो परीक्षणों में घटनाओं A, B, C में से कोई भी घटित नहीं होती। तब $1024p$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

In one trial: $P(A) = P(B) = \dfrac{1}{2}$, and C occurs when exactly one die is odd, so $P(C) = \dfrac{1}{2}$.

$A \cap B$: both odd, probability $\dfrac{1}{4}$. $A \cap C$: first odd and second even, $\dfrac{1}{4}$. $B \cap C$: $\dfrac{1}{4}$. Each equals the product of the two probabilities, so A, B, C are pairwise independent.

But $A \cap B \cap C = \phi$, since two odd numbers have an even sum. So $P(A \cap B \cap C) = 0 \neq \dfrac{1}{8}$: the events are not mutually independent, and $P(A' \cap B' \cap C')$ is not $\left(\dfrac{1}{2}\right)^3$.

By the addition theorem, $P(A \cup B \cup C) = \dfrac{3}{2} - \dfrac{3}{4} + 0 = \dfrac{3}{4}$, so $P(\text{none}) = \dfrac{1}{4}$ (directly: none occurs exactly when both dice are even).

Across the five independent trials, the number of trials in which none occurs is binomial with $n = 5$, $p_0 = \dfrac{1}{4}$. So $p = \binom{5}{2}\left(\dfrac{1}{4}\right)^2\left(\dfrac{3}{4}\right)^3 = \dfrac{10 \cdot 9 \cdot 3}{1024} = \dfrac{270}{1024}$.

Hence $1024p = 270$. (Assuming mutual independence would give success probability $\dfrac{1}{8}$ and a wrong answer.)

Final Answer: 270

Question 22

English

An urn contains 3 red balls and 3 blue balls. Three balls are drawn from the urn one at a time at random. After each draw, a red ball drawn is put back into the urn, whereas a blue ball drawn is kept aside and not put back. Then the probability that exactly two of the three balls drawn are blue is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक कलश में 3 लाल तथा 3 नीली गेंदें हैं। कलश से एक-एक करके तीन गेंदें यादृच्छया निकाली जाती हैं। प्रत्येक निकाल के बाद, निकाली गई लाल गेंद को कलश में वापस रख दिया जाता है, जबकि निकाली गई नीली गेंद को अलग रख दिया जाता है और वापस नहीं रखा जाता। तब निकाली गई तीनों गेंदों में से ठीक दो के नीली होने की प्रायिकता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A red draw is with replacement (urn unchanged); a blue draw is without replacement (one blue fewer). So the urn composition before each draw depends on how many blue balls have been drawn so far: after j blue draws it holds 3 red and $3-j$ blue balls.

Exactly two blue balls means one of the orders BBR, BRB, RBB. By the multiplication theorem:

BBR: $\frac{3}{6}\cdot\frac{2}{5}\cdot\frac{3}{4}=\frac{18}{120}=\frac{3}{20}$ (after two blue draws the urn has 3 red, 1 blue).

BRB: $\frac{3}{6}\cdot\frac{3}{5}\cdot\frac{2}{5}=\frac{18}{150}=\frac{3}{25}$ (the red draw leaves the urn at 3 red, 2 blue).

RBB: $\frac{3}{6}\cdot\frac{3}{6}\cdot\frac{2}{5}=\frac{18}{180}=\frac{1}{10}$.

The orders are mutually exclusive, so the probability is $\frac{3}{20}+\frac{3}{25}+\frac{1}{10}=0.15+0.12+0.10=0.37$.

Final Answer: 0.37

Question 23

English

In a population, a disease is present in one person out of every ten. A test for the disease gives a positive result with probability 0.9 for a person who has the disease and with probability 0.1 for a person who does not. A person chosen at random from the population is tested twice; given the disease status of the person, the results of the two tests are independent. Let p_1 be the conditional probability that the person has the disease given that the first test is positive, and let p_2 be the conditional probability that the person has the disease given that both tests are positive. Then the value of $\frac{p_2}{p_1}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक जनसंख्या में प्रत्येक दस व्यक्तियों में से एक व्यक्ति को एक रोग है। रोग के लिए एक परीक्षण, रोगी व्यक्ति के लिए प्रायिकता 0.9 से तथा रोगरहित व्यक्ति के लिए प्रायिकता 0.1 से धनात्मक परिणाम देता है। जनसंख्या से यादृच्छया चुने गए एक व्यक्ति का दो बार परीक्षण किया जाता है; व्यक्ति की रोग-स्थिति दिए होने पर दोनों परीक्षणों के परिणाम स्वतन्त्र हैं। माना कि p_1 इसकी सप्रतिबन्ध प्रायिकता है कि व्यक्ति रोगी है जबकि पहला परीक्षण धनात्मक है, तथा p_2 इसकी सप्रतिबन्ध प्रायिकता है कि व्यक्ति रोगी है जबकि दोनों परीक्षण धनात्मक हैं। तब $\frac{p_2}{p_1}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let D be 'the person has the disease', T_1, T_2 'first / second test positive'. $P(D) = 0.1$, $P(D') = 0.9$, $P(T_i \mid D) = 0.9$, $P(T_i \mid D') = 0.1$.

One test. Law of total probability: $P(T_1) = P(D)P(T_1 \mid D)+P(D')P(T_1 \mid D') = 0.09+0.09 = 0.18$.

Bayes' theorem: $p_1 = P(D \mid T_1) = \dfrac{0.09}{0.18} = 0.5$.

Two tests. Given D, the tests are independent, so $P(T_1 \cap T_2 \mid D) = 0.9^2 = 0.81$; given D', $P(T_1 \cap T_2 \mid D') = 0.1^2 = 0.01$.

Total probability: $P(T_1 \cap T_2) = 0.1 \cdot 0.81 + 0.9 \cdot 0.01 = 0.081 + 0.009 = 0.09$. Bayes: $p_2 = \dfrac{0.081}{0.09} = 0.9$.

(Equivalently, the posterior 0.5 after the first test serves as the prior for the second: $\dfrac{0.5 \cdot 0.9}{0.5 \cdot 0.9 + 0.5 \cdot 0.1} = 0.9$.)

$\dfrac{p_2}{p_1} = \dfrac{0.9}{0.5} = 1.8$. Note that the two tests are not unconditionally independent: $P(T_1 \cap T_2) = 0.09 \neq 0.18^2$.

Final Answer: 1.80

Question 24

English

A box contains n fair coins and one coin with a head on both faces. A coin is drawn at random from the box. A fair die is then rolled, and the drawn coin is tossed as many times as the number shown on the die. It is found that every toss shows a head. Given that the conditional probability that the drawn coin is the two-headed coin is 0.16, the value of n is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक डिब्बे में n अनभिनत सिक्के तथा एक ऐसा सिक्का है जिसके दोनों फलकों पर चित है। डिब्बे से एक सिक्का यादृच्छया निकाला जाता है। फिर एक अनभिनत पासा फेंका जाता है, तथा निकाले गए सिक्के को उतनी बार उछाला जाता है जितनी संख्या पासे पर आती है। पाया जाता है कि प्रत्येक उछाल में चित आता है। दिया है कि निकाले गए सिक्के के दो-चित वाला सिक्का होने की सप्रतिबन्ध प्रायिकता 0.16 है, तो n का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Tree diagram. First level: the coin, two-headed with probability $\dfrac{1}{n+1}$ and fair with probability $\dfrac{n}{n+1}$. Second level: the die shows $k \in \{1, \ldots, 6\}$ with probability $\dfrac{1}{6}$ each. Third level: k tosses.
Let E be 'every toss shows a head'. Along a two-headed branch E is certain, so $P(E \mid \text{two-headed}) = 1$. Along a fair branch with die value k, $P(E) = \dfrac{1}{2^k}$.

Law of total probability over the die: $P(E \mid \text{fair}) = \dfrac{1}{6}\left(\dfrac{1}{2} + \dfrac{1}{4} + \cdots + \dfrac{1}{64}\right) = \dfrac{1}{6} \cdot \dfrac{63}{64} = \dfrac{21}{128}$.

Bayes' theorem: $P(\text{two-headed} \mid E) = \dfrac{\frac{1}{n+1} \cdot 1}{\frac{1}{n+1} \cdot 1 + \frac{n}{n+1} \cdot \frac{21}{128}} = \dfrac{128}{128 + 21n}$.

Setting this equal to $0.16 = \dfrac{4}{25}$: $4(128 + 21n) = 25 \cdot 128 \Rightarrow 512 + 84n = 3200 \Rightarrow n = 32$.

Final Answer: 32

Question 25

English

An urn contains 3 red balls and 2 black balls. A ball is drawn at random. If it is red, it is returned to the urn together with one more red ball; if it is black, it is kept aside. A second ball is then drawn at random from the urn. Then the expected number of red balls among the two balls drawn is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक कलश में 3 लाल तथा 2 काली गेंदें हैं। एक गेंद यादृच्छया निकाली जाती है। यदि वह लाल है, तो उसे एक और लाल गेंद के साथ कलश में वापस डाल दिया जाता है; यदि वह काली है, तो उसे अलग रख दिया जाता है। फिर कलश से दूसरी गेंद यादृच्छया निकाली जाती है। तब निकाली गई दोनों गेंदों में लाल गेंदों की प्रत्याशित संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Tree diagram. First draw: red with probability $\frac{3}{5}$, black with probability $\frac{2}{5}$.

After a red first ball, the urn has 4 red and 2 black balls, so the second ball is red with probability $\frac{4}{6} = \frac{2}{3}$. After a black first ball, it has 3 red and 1 black, so the second is red with probability $\frac{3}{4}$.

Leaves: RR: $\frac{3}{5} \cdot \frac{2}{3} = \frac{2}{5}$; RB: $\frac{3}{5} \cdot \frac{1}{3} = \frac{1}{5}$; BR: $\frac{2}{5} \cdot \frac{3}{4} = \frac{3}{10}$; BB: $\frac{2}{5} \cdot \frac{1}{4} = \frac{1}{10}$ (sum 1).

Law of total probability: $P(R_2) = \frac{2}{5} + \frac{3}{10} = \frac{7}{10}$.

Let Y be the number of red balls drawn: $P(Y = 2) = \frac{2}{5}$, $P(Y = 1) = \frac{1}{5} + \frac{3}{10} = \frac{1}{2}$, $P(Y = 0) = \frac{1}{10}$.

$E(Y) = 2 \cdot \frac{2}{5} + 1 \cdot \frac{1}{2} = 1.3$, which equals $P(R_1) + P(R_2) = 0.6 + 0.7$.

Final Answer: 1.30

Question 26

English

Let $a \in (-1, 0)$. A point (x, y) is chosen at random in the rectangle $R = \{(x, y) : 0 \le x \le 2,\ -1 \le y \le 4\}$, all points of R being equally likely. The probability that the chosen point lies between the curve $y = x^2 + a$ and the x-axis, that is, that y lies between 0 and $x^2 + a$, is $\frac{7}{30}$. Then the value of a is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $a \in (-1, 0)$ है। आयत $R = \{(x, y) : 0 \le x \le 2,\ -1 \le y \le 4\}$ में एक बिन्दु (x, y) यादृच्छया चुना जाता है, R के सभी बिन्दु समसम्भाव्य हैं। चुने गए बिन्दु के वक्र $y = x^2 + a$ तथा x-अक्ष के बीच स्थित होने की, अर्थात् y के 0 तथा $x^2 + a$ के बीच होने की, प्रायिकता $\frac{7}{30}$ है। तब a का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Geometric probability: $P = \dfrac{\text{favourable area}}{\text{area of } R}$, and the area of R is $2 \cdot 5 = 10$. So the favourable area is $10 \cdot \dfrac{7}{30} = \dfrac{7}{3}$.

For $a \in (-1, 0)$ and $0 \le x \le 2$, $x^2 + a$ lies in $[a, 4 + a] \subset (-1, 4)$, so the whole region between the curve and the x-axis lies inside R. Its area is $\int_0^2 |x^2 + a|\, dx$, not $\int_0^2 (x^2 + a)\, dx$, because the curve crosses the axis.

Let $a = -c^2$ with $c \in (0, 1)$. Then $x^2 + a < 0$ on $(0, c)$ and > 0 on $(c, 2)$.

$\int_0^c (c^2 - x^2)\, dx = c^3 - \dfrac{c^3}{3} = \dfrac{2c^3}{3}$ and $\int_c^2 (x^2 - c^2)\, dx = \dfrac{8 - c^3}{3} - c^2(2 - c) = \dfrac{8}{3} - 2c^2 + \dfrac{2c^3}{3}$.

Total area $= \dfrac{4c^3}{3} - 2c^2 + \dfrac{8}{3} = \dfrac{7}{3} \Rightarrow 4c^3 - 6c^2 + 1 = 0$. Since $c = \dfrac{1}{2}$ is a root, $4c^3 - 6c^2 + 1 = (2c - 1)(2c^2 - 2c - 1)$, and the other roots $\dfrac{1 \pm \sqrt{3}}{2}$ are not in $(0, 1)$.

So $c = \dfrac{1}{2}$ and $a = -\dfrac{1}{4} = -0.25$. (Ignoring the sign change, $\dfrac{8}{3} + 2a = \dfrac{7}{3}$ would give the wrong value $a = -\dfrac{1}{6}$.)

Final Answer: -0.25

Question 27

English

A discrete random variable X takes only the values -3, 0 and 2, each with a positive probability. The mean of X is 0 and the variance of X is 3. Then the mean of the random variable X^3 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक असतत यादृच्छिक चर X केवल मान -3, 0 तथा 2 लेता है, प्रत्येक धनात्मक प्रायिकता के साथ। X का माध्य 0 तथा X का प्रसरण 3 है। तब यादृच्छिक चर X^3 का माध्य _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Probability distribution: $P(X=-3)=\alpha$, $P(X=0)=\beta$, $P(X=2)=\gamma$ with $\alpha,\beta,\gamma>0$ and $\alpha+\beta+\gamma=1$.

Mean: $E(X)=-3\alpha+0\cdot\beta+2\gamma=0$, so $\gamma=\dfrac{3\alpha}{2}$.

Variance: $\mathrm{Var}(X)=E(X^2)-[E(X)]^2=E(X^2)=9\alpha+4\gamma=9\alpha+6\alpha=15\alpha=3$, so $\alpha=\dfrac{1}{5}$, $\gamma=\dfrac{3}{10}$ and $\beta=1-\dfrac{1}{5}-\dfrac{3}{10}=\dfrac{1}{2}$. All are positive, as required.

Mean of X^3: $E(X^3)=(-27)\cdot\dfrac{1}{5}+0+8\cdot\dfrac{3}{10}=-5.4+2.4=-3$.

Final Answer: -3

Question 28

English

Ten observations $x_1, x_2, \ldots, x_{10}$ satisfy $\sum_{i=1}^{10} x_i = 50$ and $\sum_{i=1}^{10} x_i^2 = 340$. Two of the ten observations are chosen at random without replacement, all $\binom{10}{2}$ choices being equally likely, and D denotes the square of the difference of the two chosen observations. Then the mean of D is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

दस प्रेक्षण $x_1, x_2, \ldots, x_{10}$ इस प्रकार हैं कि $\sum_{i=1}^{10} x_i = 50$ तथा $\sum_{i=1}^{10} x_i^2 = 340$ है। इन दस प्रेक्षणों में से दो प्रेक्षण बिना प्रतिस्थापन के यादृच्छया चुने जाते हैं, सभी $\binom{10}{2}$ चयन समसम्भाव्य हैं, तथा D चुने गए दोनों प्रेक्षणों के अन्तर का वर्ग है। तब D का माध्य _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Variance of the ungrouped data: $\bar{x}=\dfrac{50}{10}=5$ and $\sigma^2=\dfrac{1}{10}\sum x_i^2-\bar{x}^2=34-25=9$.

D is a discrete random variable taking the value $(x_i-x_j)^2$ for each of the 45 pairs $i<j$ with probability $\dfrac{1}{45}$, so $E(D)=\dfrac{1}{45}\sum_{i<j}(x_i-x_j)^2$.

Double sum: $\sum_{i=1}^{n}\sum_{j=1}^{n}(x_i-x_j)^2=\sum_{i,j}(x_i^2+x_j^2-2x_ix_j)=2n\sum x_i^2-2\left(\sum x_i\right)^2$. The terms with $i=j$ vanish and each pair $i<j$ appears twice, so $\sum_{i<j}(x_i-x_j)^2=n\sum x_i^2-\left(\sum x_i\right)^2=n^2\sigma^2$.

Here $\sum_{i<j}(x_i-x_j)^2=10\cdot 340-50^2=900$ $(=100\cdot 9)$.

$E(D) = \dfrac{900}{45} = 20$. In general $E(D) = \dfrac{n^2\sigma^2}{\binom{n}{2}} = \dfrac{2n\sigma^2}{n-1}$, slightly more than $2\sigma^2 = 18$ because the two observations are distinct. (Example data: five values 2 and five values 8.)

Final Answer: 20

Question 29

English

Let X denote the number of successes in 5 independent trials, each with the same probability of success p, where $0 < p < 1$. It is given that the variance of X is 1.2 and that $P(X = 4) > P(X = 1)$. Then the value of $\dfrac{P(X = 5)}{P(X = 4)}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि X, 5 स्वतन्त्र परीक्षणों में सफलताओं की संख्या है, प्रत्येक परीक्षण में सफलता की प्रायिकता समान p है, जहाँ $0 < p < 1$ है। दिया है कि X का प्रसरण 1.2 है तथा $P(X = 4) > P(X = 1)$ है। तब $\dfrac{P(X = 5)}{P(X = 4)}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$X \sim B(5, p)$ with $q = 1 - p$. Variance of a binomial distribution: $npq = 5pq = 1.2$, so $pq = 0.24$, i.e. $p^2 - p + 0.24 = 0$, giving $p = 0.6$ or $p = 0.4$.

Choosing the root: $P(X = 4) = \binom{5}{4}p^4q$ and $P(X = 1) = \binom{5}{1}pq^4$, so $\dfrac{P(X = 4)}{P(X = 1)} = \left(\dfrac{p}{q}\right)^3$. This exceeds 1 only if $p > q$, so $p = 0.6$, $q = 0.4$ (mean 3, variance 1.2).

$$\frac{P(X = 5)}{P(X = 4)} = \frac{p^5}{5p^4q} = \frac{p}{5q} = \frac{0.6}{5 \cdot 0.4} = 0.3.$$

Final Answer: 0.30

Question 30

English

In one round, three letters are placed at random into three envelopes addressed to their three recipients, one letter in each envelope, all 3! placements being equally likely. A round is called a miss if no letter is placed in its correctly addressed envelope. Six rounds are performed independently. The probability that no two consecutive rounds are both misses is $\frac{k}{3^6}$. Then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक चक्र में तीन पत्र, उनके तीन प्राप्तकर्ताओं के पते लिखे तीन लिफाफों में यादृच्छया रखे जाते हैं, प्रत्येक लिफाफे में एक पत्र, तथा सभी $3!$ व्यवस्थाएँ समसम्भाव्य हैं। एक चक्र को चूक कहा जाता है यदि कोई भी पत्र अपने सही पते वाले लिफाफे में नहीं रखा जाता। छह चक्र स्वतन्त्र रूप से किए जाते हैं। कोई भी दो क्रमागत चक्र दोनों चूक न होने की प्रायिकता $\frac{k}{3^6}$ है। तब k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

One round: of the $3! = 6$ placements, the derangements are the two cyclic placements, so $D_3 = 3!\left(1 - 1 + \frac{1}{2!} - \frac{1}{3!}\right) = 2$ and $P(\text{miss}) = \frac{1}{3}$, $P(\text{not a miss}) = \frac{2}{3}$. Rounds are independent.

Recursion: let b_n be the probability that among n rounds no two consecutive rounds are both misses ($b_0 = b_1 = 1$). Condition on the first round. If it is not a miss (probability $\frac{2}{3}$), the remaining $n-1$ rounds must have the property: $\frac{2}{3}b_{n-1}$. If it is a miss (probability $\frac{1}{3}$), the second round must not be a miss (probability $\frac{2}{3}$) and the remaining $n-2$ rounds must have the property: $\frac{2}{9}b_{n-2}$.

So $b_n = \frac{2}{3}b_{n-1} + \frac{2}{9}b_{n-2}$. Multiplying by 3^n and writing $c_n = 3^n b_n$: $c_n = 2c_{n-1} + 2c_{n-2}$ with $c_0 = 1$, $c_1 = 3$.

$c_2 = 2 \cdot 3 + 2 \cdot 1 = 8$ (check: $1 - \frac{1}{9} = \frac{8}{9}$), $c_3 = 16 + 6 = 22$, $c_4 = 44 + 16 = 60$, $c_5 = 120 + 44 = 164$, $c_6 = 328 + 120 = 448$.

Hence the probability is $\frac{448}{729}$ and $k = 448$.

Final Answer: 448